Logic and Proof

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Introduction

A *proposition* is a declarative sentence with a definite truth value in $\{\text{True}, \text{False}\}$. A *proof* is a finite chain of propositions, each justified by an axiom or a prior theorem under valid rules of inference. Every theoretical claim in mathematics — and every formal guarantee in a machine-learning paper — ultimately reduces to such a chain. The goal of this lesson is to make the vocabulary precise and to drill the four basic proof strategies.

Definitions

Definition 1 (Proposition). *A* proposition *is a statement that is either true or false (but not both). Variables p, q, r range over propositions.*

Definition 2 (Logical connectives). *For propositions p, q we define:*

$$\begin{aligned} \neg p &: \text{“not } p\text{”}, \quad \text{true iff } p \text{ is false}; \\ p \wedge q &: \text{“}p \text{ and } q\text{”}, \quad \text{true iff both are true}; \\ p \vee q &: \text{“}p \text{ or } q\text{”}, \quad \text{true iff at least one is true}; \\ p \Rightarrow q &: \text{“if } p \text{ then } q\text{”}, \quad \text{false iff } p \text{ true and } q \text{ false}; \\ p \Leftrightarrow q &: \text{“}p \text{ iff } q\text{”}, \quad \text{true iff } p \text{ and } q \text{ have the same value.} \end{aligned}$$

Definition 3 (Tautology). *A compound proposition is a* tautology *if it is true under every assignment of truth values to its atomic variables.*

Definition 4 (Predicate, quantifiers). *A* predicate $P(x)$ *on a domain D is a statement whose truth value depends on $x \in D$. The* universal quantifier $\forall x \in D,\ P(x)$ *asserts P holds for every x; the* existential quantifier $\exists x \in D,\ P(x)$ *asserts P holds for at least one x.*

Properties and Theorems

Proposition 1 (Contrapositive equivalence). *For all propositions p, q, $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$.*

Proposition 2 (De Morgan). *$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ and $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$. For quantifiers, $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$ and $\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$.*

Theorem 1 (Principle of Mathematical Induction). *Let $P(n)$ be a predicate on $\mathbb{N} = \{1, 2, 3, \dots\}$. Suppose*

1. *(Base case) $P(1)$ is true, and*
2. *(Inductive step) for every $k \in \mathbb{N}$, $P(k) \Rightarrow P(k+1)$.*

Then $P(n)$ is true for every $n \in \mathbb{N}$.

Worked Examples

Direct proof

Proposition 3. *If n is an even integer, then n^2 is even.*

Proof. By hypothesis $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is twice an integer, hence even. □

Contrapositive

Proposition 4. *If n^2 is an even integer, then n is even.*

Proof. We prove the contrapositive: if n is odd, then n^2 is odd. Write $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. □

Contradiction

Proposition 5. *$\sqrt{2}$ is irrational.*

Proof. Suppose for contradiction $\sqrt{2} = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$, $\gcd(a, b) = 1$. Then $a^2 = 2b^2$, so a^2 is even, so a is even (contrapositive lemma above). Write $a = 2c$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$, so b is even. But then $\gcd(a, b) \geq 2$, contradicting $\gcd(a, b) = 1$. □

Induction

Theorem 2 (Closed form for the first n positive integers)**.** *For every $n \geq 1$,*

$$\sum_{k=1}^{n} k \;=\; \frac{n(n+1)}{2}.$$

Proof. We argue by induction on n, applying Theorem 1.

Base case. For $n = 1$ the left-hand side is $\sum_{k=1}^{1} k = 1$ and the right-hand side is $1 \cdot 2/2 = 1$. Both equal 1.

Inductive step. Fix $m \geq 1$ and assume the inductive hypothesis

$$\sum_{k=1}^{m} k = \frac{m(m+1)}{2}. \tag{IH}$$

We must show the statement for $n = m + 1$. Compute

$$\begin{aligned} \sum_{k=1}^{m+1} k &= \left(\sum_{k=1}^{m} k\right) + (m+1) \\ &\overset{\text{(IH)}}{=} \frac{m(m+1)}{2} + (m+1) \\ &= (m+1)\left(\frac{m}{2} + 1\right) \\ &= (m+1) \cdot \frac{m+2}{2} \\ &= \frac{(m+1)\big((m+1)+1\big)}{2}, \end{aligned}$$

which is the claim for $n = m + 1$.

By Theorem 1, the identity holds for every $n \geq 1$. ☐

Connection to ML

Every formal guarantee in a machine-learning paper is a quantified proposition: a convergence rate is $\forall \epsilon > 0$, $\exists N$, $\forall n \geq N$, $\|\theta_n - \theta^*\| < \epsilon$; a generalisation bound is $\forall \delta, \Pr[\,\text{risk gap} \leq f(\delta)\,] \geq 1 - \delta$. Identifying the proof strategy — direct construction, contrapositive, contradiction, or induction (often on network depth or training step) — is the first step in critiquing the paper. For example, a Lipschitz bound for a deep network is classically proved by induction on the layer index, with the inductive hypothesis that the partial network up to layer k has Lipschitz constant $\prod_{i \leq k} L_i$.

Exercises

1. (Direct.) Prove that the sum of two odd integers is even.
2. (Contradiction.) Prove that there is no largest prime. *Hint:* if $p_1, \ldots, p_n$ were all the primes, consider $N = p_1 \cdots p_n + 1$.
3. (Induction.) Prove that $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ for every $n \geq 1$.
4. (Truth tables.) Verify by truth table that $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$ is a tautology.